

Question 1 – ECDF of Student Marks

```
marks <- c(45,52,58,60,65,68,72,75,80,90)

ecdf_marks <- ecdf(marks)

ecdf_values <- ecdf_marks(marks)

data.frame(Marks=marks, Cumulative_Probability=ecdf_values)

plot(ecdf_marks,

     main="ECDF of Student Marks",

     xlab="Marks",

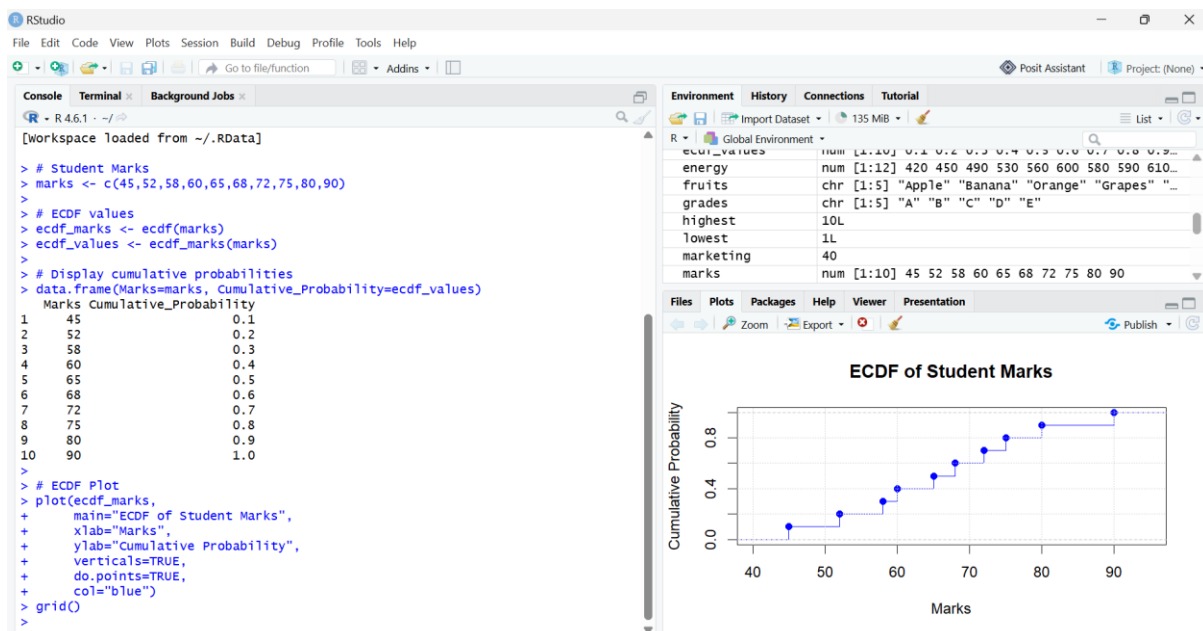
     ylab="Cumulative Probability",

     verticals=TRUE,

     do.points=TRUE,

     col="blue")

grid()
```



Question 2 – ECDF of Daily Steps

```
steps <- c(4200,5000,5600,6100,6800,7200,8100,9200,10400,12000)

ecdf_steps <- ecdf(steps)

plot(ecdf_steps,

     main="ECDF of Daily Steps",
```

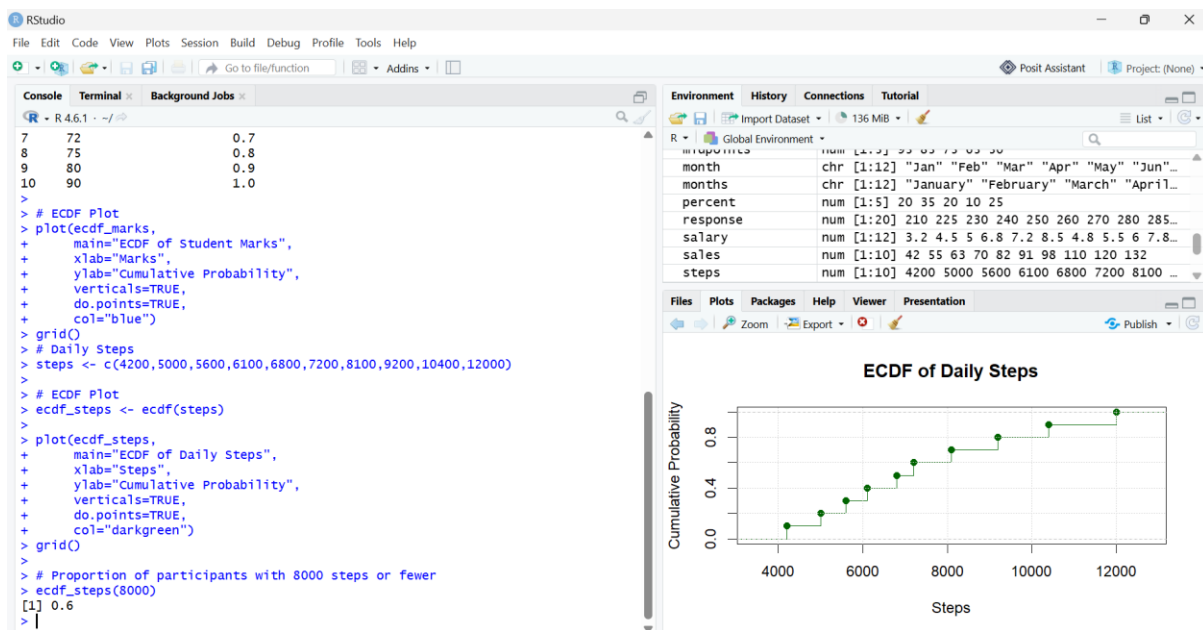
```

xlab="Steps",
ylab="Cumulative Probability",
verticals=TRUE,
do.points=TRUE,
col="darkgreen")

grid()

ecdf_steps(8000)

```



Question 3 – Histogram and Density Plot

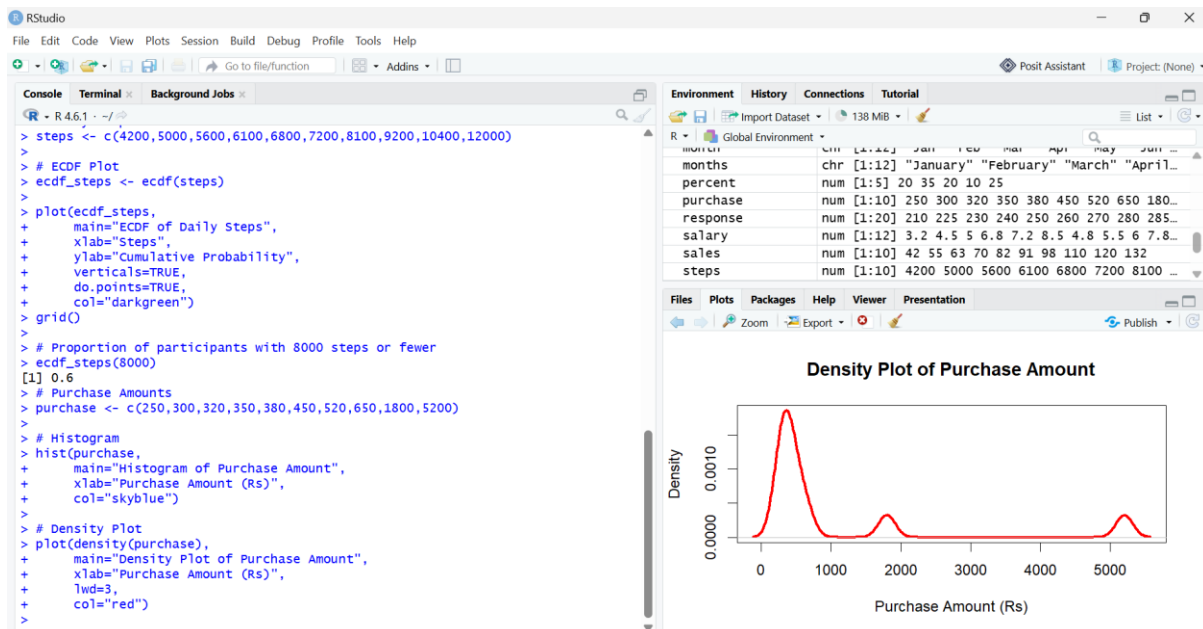
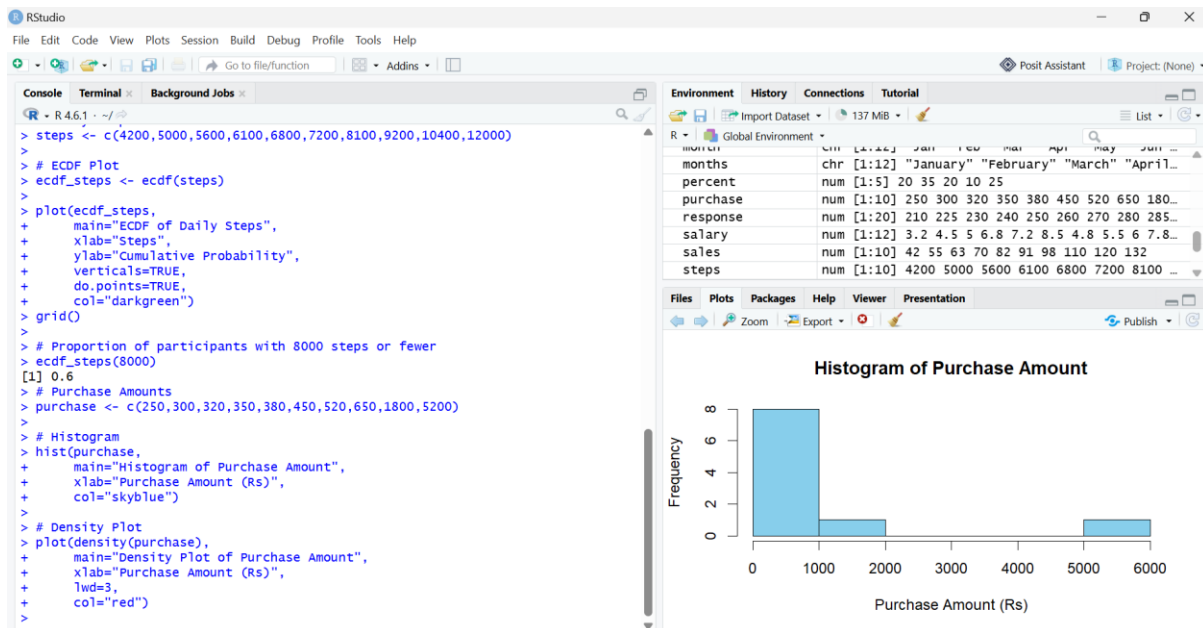
```
purchase <- c(250,300,320,350,380,450,520,650,1800,5200)
```

```

hist(purchase,
     main="Histogram of Purchase Amount",
     xlab="Purchase Amount (Rs)",
     col="skyblue")

plot(density(purchase),
     main="Density Plot of Purchase Amount",
     xlab="Purchase Amount (Rs)",
     lwd=3,
     col="red")

```



Question 4 – Normal Q-Q Plot

```
waiting <- c(12,15,18,20,22,24,25,28,30,32)
```

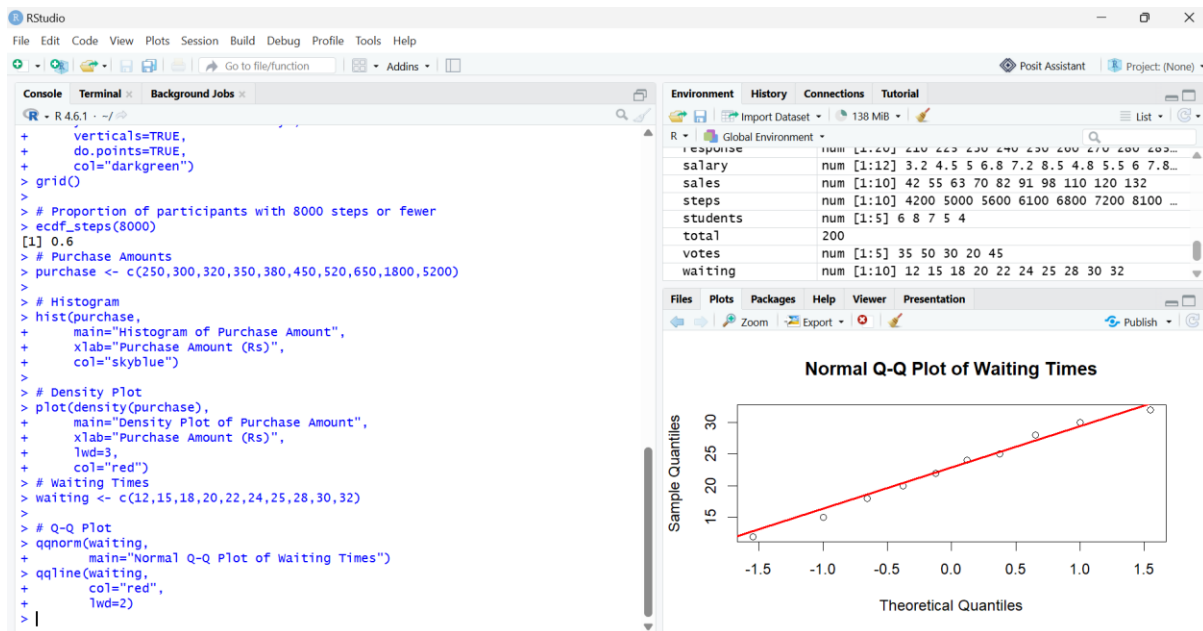
```
qqnorm(waiting,
```

```
      main="Normal Q-Q Plot of Waiting Times")
```

```
qqline(waiting,
```

```
      col="red",
```

```
      lwd=2)
```



Question 5 – Histogram and Q-Q Plot of Monthly Income

```
income <- c(18000,22000,24000,26000,28000,
            30000,34000,42000,85000,200000)
```

```
hist(income,
     main="Histogram of Monthly Income",
     xlab="Income (Rs)",
     col="lightblue")

qqnorm(income,
       main="Normal Q-Q Plot of Monthly Income")

qqline(income,
       col="red",
       lwd=2)
```

